

WASP-43b

R-0648

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_170311 · created 2026-07-25T21:11:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457822.8667 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.056
Transit depth (Rp/Rs)^2	2.43 +/- 1.76 [%]
Orbital Inclination (inc)	85.57 +/- 2.94
Airmass coefficient 1 (a1)	0.346 +/- 0.06
Airmass coefficient 2 (a2)	0.4 +/- 0.41
Scatter in the residuals of the lightcurve fit is	15.94 %
Best Comparison Star	#1 - [517, 153]
Optimal Aperture	1.22
Optimal Annulus	3.0
Transit Duration (day)	0.0 +/- 0.0

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.156 $\pm$ 0.056 vs 0.1594 (deviation 0.06 σ)
Duration fit vs published	0.0 d vs 0.0512 d (deviation inf σ)
Mid-time O-C	-1169.8 min
Depth SNR	2.8
Residual scatter	15.94 %

Light curve

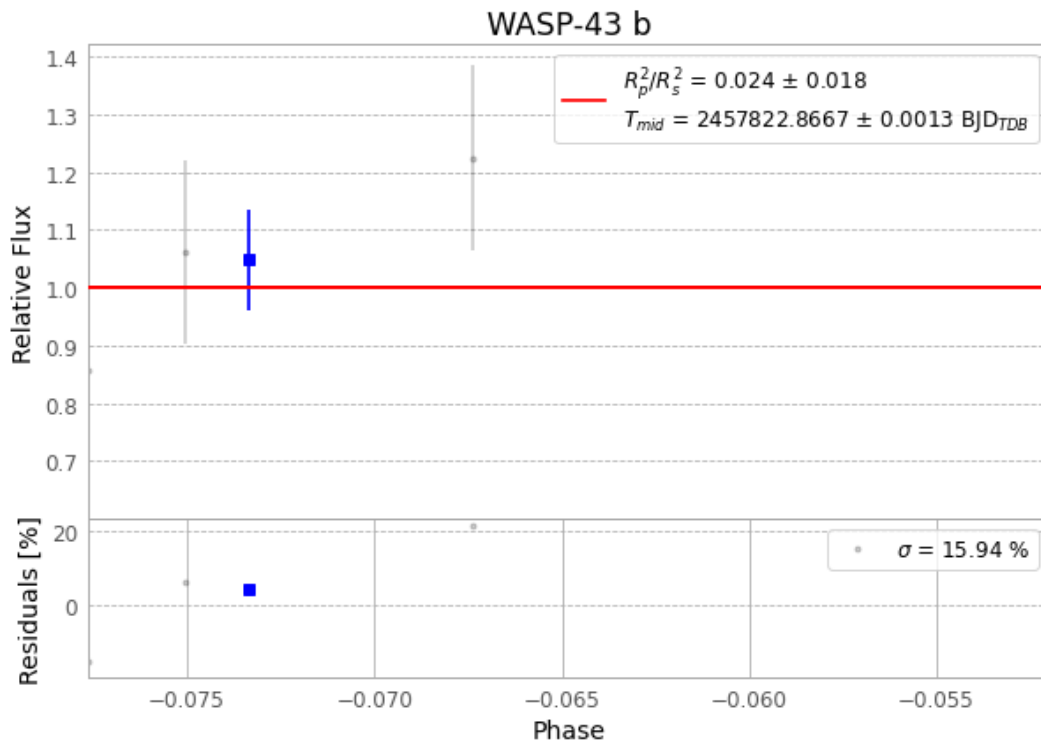


Figure 1 — FinalLightCurve_WASP-43 b_2017-03-10.png (SHA-256 64941f43a23f8f91...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [251, 337], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d248e6cf87eb433...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

81 FITS frames (53 MB), WASP-43170311022712.FITS → WASP-43170311063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170311.log (93ae1e1f1d8c...), FinalLightCurve_WASP-43 b_2017-03-10.png (64941f43a23f...), FinalLightCurve_WASP-43 b_2017-03-10.csv (730286f8c71f...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 21:11Z.